Perfect 👍
Chandan, ***Robotics_WiFi – Wi-Fi LED & Buzzer Control Panel*** project beginner–friendly hai aur BCA + Robotics dono ke liye mast demo project banega.
Main **from scratch, step-by-step** bata raha hoon — **components → connections → code → app/web control → testing**.

◆ PROJECT OVERVIEW

Goal:
Mobile / Laptop se **Wi-Fi ke through LED aur Buzzer ON/OFF control** karna.

Concepts used:

- IoT
- Wi-Fi communication
- Web Server
- GPIO control

◆ REQUIRED COMPONENTS

1. **ESP8266 NodeMCU** (Best)
   (ya ESP32 bhi use kar sakte ho)
2. **LED** (any color)
3. **220Ω resistor**
4. **Buzzer** (Active buzzer recommended)
5. **Breadboard**
6. **Jumper wires**
7. **USB cable**
8. **Mobile / Laptop (Wi-Fi enabled)**

◆ STEP 1: SOFTWARE SETUP (Very Important)

✅ 1. Arduino IDE Install

- Download: https://www.arduino.cc/en/software

✅ 2. ESP8266 Board Add

Arduino IDE →
File → Preferences → Additional Boards URL

```
https://arduino.esp8266.com/stable/package_esp8266com_index.json
```

Then:

- Tools → Board → Board Manager
- Search **ESP8266**
- Install

✅ 3. Select Board

```
Board: NodeMCU 1.0 (ESP-12E Module)
Port: COMx
```

◆ STEP 2: CIRCUIT CONNECTIONS

🔌 LED Connection

LED	NodeMCU
+ (Long leg)	D1
–	220Ω resistor → GND

🔊 Buzzer Connection

Buzzer	NodeMCU
+	D2
–	GND

⚠ **Note:**

- External power NOT required
- NodeMCU USB se hi power milegi

◆ STEP 3: LOGIC WORKING

1. ESP8266 Wi-Fi se connect hota hai
2. Apna **web server** banata hai
3. Browser me IP open karte hi **Control Panel** dikhega

4. Button click → LED / Buzzer ON/OFF

◆ STEP 4: FULL CODE (COPY–PASTE)

```
#include <ESP8266WiFi.h>

const char* ssid = "YOUR_WIFI_NAME";
const char* password = "YOUR_WIFI_PASSWORD";

WiFiServer server(80);

int ledPin = D1;
int buzzerPin = D2;

void setup() {
  Serial.begin(9600);

  pinMode(ledPin, OUTPUT);
  pinMode(buzzerPin, OUTPUT);

  digitalWrite(ledPin, LOW);
  digitalWrite(buzzerPin, LOW);

  WiFi.begin(ssid, password);
  Serial.print("Connecting");

  while (WiFi.status() != WL_CONNECTED) {
    delay(500);
    Serial.print(".");
  }

  Serial.println("\nWiFi Connected");
  Serial.println(WiFi.localIP());

  server.begin();
}

void loop() {
  WiFiClient client = server.available();
  if (!client) return;

  String request = client.readStringUntil('\r');
  client.flush();

  if (request.indexOf("/LED=ON") != -1) digitalWrite(ledPin, HIGH);
  if (request.indexOf("/LED=OFF") != -1) digitalWrite(ledPin, LOW);
  if (request.indexOf("/BUZZER=ON") != -1) digitalWrite(buzzerPin, HIGH);
  if (request.indexOf("/BUZZER=OFF") != -1) digitalWrite(buzzerPin, LOW);

  client.println("HTTP/1.1 200 OK");
  client.println("Content-Type: text/html");
  client.println("");
```

```
  client.println("<html><body><center>");
  client.println("<h2>WiFi LED & Buzzer Control</h2>");
  client.println("<a href='/LED=ON'><button>LED ON</button></a>");
  client.println("<a href='/LED=OFF'><button>LED OFF</button></a><br><br>");
  client.println("<a href='/BUZZER=ON'><button>BUZZER ON</button></a>");
  client.println("<a href='/BUZZER=OFF'><button>BUZZER OFF</button></a>");
  client.println("</center></body></html>");
}
```

● **Wi-Fi name & password change karna mat bhoolna**

◆ STEP 5: UPLOAD & TEST

1. Code upload karo
2. Serial Monitor open karo (9600 baud)
3. **IP address note karo**
4. Browser me type karo:

```
http://IP_ADDRESS
```

🎉 **Control panel open ho jayega**

◆ STEP 6: PROJECT ENHANCEMENTS (For BCA Project)

✅ Mobile App (MIT App Inventor)
✅ Password protected control
✅ Relay add karke AC device control
✅ Firebase / Blynk
✅ Stylish HTML + CSS UI

◆ PROJECT TITLE (Report ke liye)

“Robotics Wi-Fi Based LED & Buzzer Control Panel using ESP8266”

Agar chaho to main:

- 📄 **Project report (BCA format)**
- 📱 **Android app**
- 📷 **Circuit diagram**
- 📹 **YouTube demo script**

bhi bana deta hoon 😉
Bas batao **next kya chahiye?**